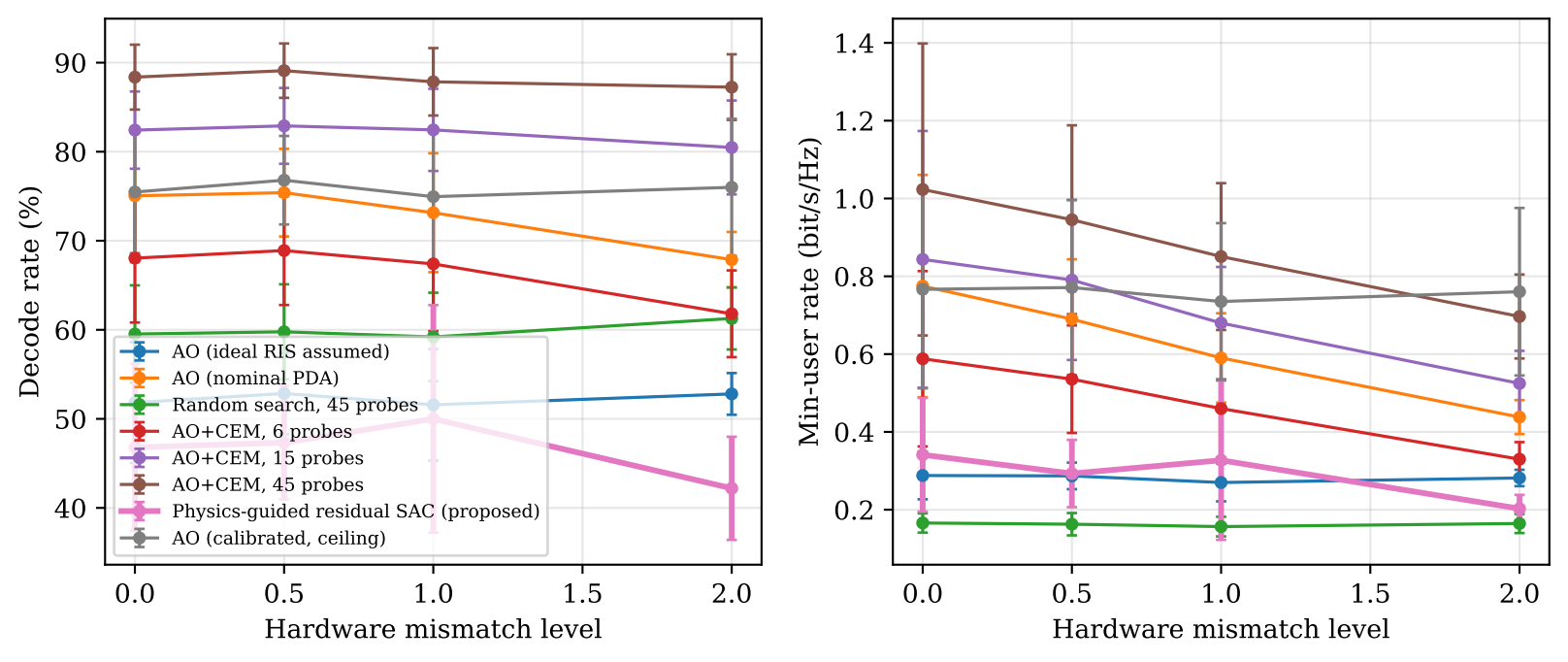
Decode rate (%)
90
80
70
60
50
40
0.0
0.5
1.0
1.5
2.0
Hardware mismatch level
AO (ideal RIS assumed)
AO (nominal PDA)
Random search, 45 probes
AO+CEM, 6 probes
AO+CEM, 15 probes
AO+CEM, 45 probes
Physics-guided residual SAC (proposed)
AO (calibrated, ceiling)
Min-user rate (bit/s/Hz)
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.5
1.0
1.5
2.0
Hardware mismatch level